

Nibedita Sahu

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with a strong mathematical foundation, focused on building end-to-end ML systems and AI-powered decision tools. Skilled in translating statistical and analytical concepts into practical Python and SQL workflows, with experience in predictive modeling, decision intelligence, and technical communication.

SKILLS

- **Data Science & ML:** Mathematical Modeling, Supervised Learning, Time-Series Forecasting using Python (NumPy, Pandas, Scikit-learn, XGBoost, StatsModels, SciPy, DuckDB) and SQL (MySQL, PostgreSQL).
- **AI & Decision Intelligence:** Generative AI (LLMs), AI Integration, Decision Intelligence, and translating analytical outputs into actionable business insights and data-driven decisions.
- **Visual Storytelling:** Data Visualization, EDA, and Data Storytelling using Matplotlib, Seaborn, Power BI, and Tableau to communicate insights clearly to technical and non-technical stakeholders.

EXPERIENCE

Independent Data Science Researcher & Technical Writer | Medium (May 2023 – Present)

- Curate specialized research lists including [ML for DS](#), [Intro to Statistics](#), [NumPy for DS & DA](#), and [Pandas for DS & DA](#), with articles featured in prominent Data Science publications.
- Maintain [Data Projects](#) and [DS Playground](#) lists, providing end-to-end walkthroughs of Machine Learning systems and exploratory lab projects.

Technical Content Writer | GeeksforGeeks (Sep 2023 – Sep 2024)

- Authored 10+ technical articles on SQL Optimization and Python Performance, clearing a rigorous editorial review process for technical accuracy and depth.
- Translated complex programming and data-related concepts into clear, practical technical content for a developer audience.

PROJECTS

Airline Payment Fraud Decision Intelligence System

Python (NumPy, Pandas, Scikit-learn, XGBoost, SHAP, DuckDB, Streamlit)

- Built an end-to-end fraud decision intelligence system to predict transaction-level fraud risk for airline payment transactions.
- Developed an XGBoost model and applied explainable AI to identify and communicate the factors influencing individual fraud predictions.
- Translated predicted risk into business-oriented payment recommendations and deployed the system as an interactive Streamlit application.

Customer Retention & Revenue Optimization System

Python (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn) | SQL | Generative AI (LLMs)

- Built an end-to-end system to predict customer churn and purchase probability, enabling targeted strategies based on customer behavior.
- Applied predictive modeling and optimization techniques to identify high-value customers under budget constraints, focusing on measurable business impact.
- Integrated a Generative AI layer to translate model outputs into business insights, improving decision-making and interpretability.

End-to-End Time Series Analysis & Forecasting (Nutrition Data)

Python (NumPy, Pandas, Matplotlib, StatsModels, Scikit-learn) | Power BI | Tableau

- Conducted end-to-end time-series analysis to identify trends, seasonality, and consumption patterns while evaluating statistical assumptions.
- Developed and compared forecasting approaches with emphasis on model performance, interpretability, and reliable predictions.
- Communicated analytical and forecasting insights through interactive Power BI and Tableau dashboards.

EDUCATION

Bachelor of Science in Mathematics

Kalahandi University, Bhawanipatna | Odisha, India | 2020 - 2023